MOHAMED DILROSE P M

Data Scientist / Data Analyst

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LinkedIn | Portfolio | GitHub

PROFESSIONAL SUMMARY

Data Scientist & Analyst with an M.Sc. in Statistics and a strong foundation in statistical inference, predictive modeling, and business intelligence. Skilled in **SQL**, **Python**, **Power BI**, and **Excel**, with hands-on experience building end-to-end analytics pipelines—from data wrangling and ETL to machine learning (regression, classification, clustering) and dashboard development. Adept at transforming complex datasets into decision-ready insights that support strategic and operational goals.

EDUCATION

- **Pondicherry University** *September 2023 - May 2025*
  M.Sc. Statistics | CGPA: 7.38/10.00 Pondicherry, India
- **University of Calicut** *September 2020 - March 2023*
  B.Sc. Statistics Thrissur, India

PROJECTS

- **360 Retail Analytics: Customer Segmentation & Sales Insights – End-to-End Project** *July 2025*
  Tools: Excel, SQL (MySQL), Python (Pandas, Scikit-learn), Power BI [GitHub]
  - Analyzed **10,000+** retail transactions to uncover sales patterns, seasonal trends, and customer purchasing behavior for targeted marketing.
  - Cleaned and transformed raw data in Excel, ensuring high-quality datasets for advanced SQL and clustering analysis.
  - Leveraged advanced **SQL (joins, CTEs, window functions)** to analyze monthly sales trends and customer purchase patterns, uncovering seasonality and category-level performance drivers.
  - Performed customer segmentation using **RFM analysis** and **KMeans clustering** to classify customers into **3 key segments**, revealing that the **top 30% of customers contributed 60% of revenue**.
  - Designed a multi-page interactive Power BI dashboard to visualize sales trends, customer segments, and regional performance, enabling stakeholders to identify underperforming regions and seasonal demand spikes.

- **HR Analytics – Predicting Employee Attrition and Retention Drivers** *July 2025*
  Tools: Excel, SQL (MySQL), Python (Pandas, Scikit-learn), Power BI [GitHub]
  - Conducted a root-cause analysis of the IBM HR Analytics dataset **(1,470 records)**, identifying key drivers of employee attrition and informing proactive retention strategies.
  - Performed data cleaning and EDA using Excel, SQL, and Python to uncover key attrition drivers across demographics, job performance, and compensation metrics.
  - Built and tuned multiple ML models **(Logistic Regression, Random Forest, XGBoost)** to forecast employee attrition, with **SMOTE** balancing and **RandomizedSearchCV** tuning, improving recall for high-risk attrition cases by **27%**.
  - Achieved the best performance with **Logistic Regression** (**86% accuracy**, **0.82 ROC-AUC**, high recall), providing a robust framework to predict likely leavers.
  - Designed an interactive Power BI dashboard visualizing attrition patterns, risk segments, job roles, and compensation trends to enable targeted HR retention planning.

- **Hybrid Supervised & Anomaly Detection Model for Financial Fraud** *September 2025*
  Tools: Python (Pandas, Numpy, Scikit-learn, Matplotlib, Seaborn) [GitHub]
  - Built a fraud detection pipeline on 284K+ credit card transactions with only **0.17%** fraud rate, handling severe class imbalance using SMOTE oversampling.
  - Developed and compared supervised models **(Logistic Regression, Random Forest, XGBoost)** and unsupervised anomaly detectors **(Isolation Forest, One-Class SVM, LOF)**, achieving ROC-AUC up to **0.98**.
  - Designed a hybrid ensemble model combining **Random Forest with One-Class SVM**, improving recall to **86%** while maintaining **79%** precision (F1 = **0.82**), reducing missed fraud cases.
  - Optimized decision thresholds and implemented a probability boosting strategy to enhance sensitivity to rare fraud patterns without over-flagging legitimate transactions.

CERTIFICATIONS

- IBM Data Science Professional Certificate — IBM *September 2025*
- Data Science: Machine Learning — Harvard University (edX) *May 2025*
- Deep Learning Specialization — DeepLearning.AI *October 2025*
- Data Analytics with Python — NPTEL *April 2024*
- Advanced Microsoft Power BI — Coursera *June 2025*
- Master SQL for Data Science — Udemy *June 2025*

SKILLS

- **Programming Languages:** **Python** (Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch), **R**
- **Data Analysis & Databases:** **SQL** (MySQL - Joins, CTEs, Window functions), **MS Excel** (Pivot tables, Advanced formulas, VBA), Google Sheets
- **Machine Learning:** Regression, Classification, Clustering, Anomaly detection, Time Series Forecasting, Feature Engineering, Model Evaluation, Hyperparameter Tuning
- **Data Visualization:** **Power BI** (DAX, KPI dashboards), **Tableau**, Matplotlib, Seaborn
- **Statistical Analysis:** Exploratory Data Analysis (EDA), Hypothesis Testing (A/B Testing, t-tests, ANOVA), Experimental Design, Correlation Analysis
- **Tools & Platforms:** Git/GitHub, Streamlit & Gradio (Deployment), Jupyter Notebook, VS Code

ADDITIONAL INFORMATION

Languages: **English** (Proficient) | **Hindi** (Proficient) | **Malayalam** (Native)